CF04

① $S = \max(|W|)/127$
$= |-2.31|/127$
$= 0.01819$

② $W_q = \text{round}(W/S)$

$$\begin{bmatrix} 47 & -66 & 19 & 115 \\ -4 & 50 & -103 & 7 \\ 85 & 2 & -24 & -127 \\ -10 & 57 & 42 & 30 \end{bmatrix}$$

③ $W_deq = W_q \times S$

$$\begin{bmatrix} 0.85 & -1.20 & 0.35 & 2.09 \\ -0.07 & 0.91 & -1.87 & 0.13 \\ 1.55 & 0.04 & -0.44 & -2.31 \\ -0.18 & 1.04 & 0.76 & 0.55 \end{bmatrix}$$

④ $$\begin{bmatrix} 0.00488 & 0.00047 & 0.00559 & 0.00827 \\ 0.00276 & 0.00055 & 0.00654 & 0.00732 \\ 0.00394 & 0.00638 & 0.00346 & 0.00000 \\ 0.00189 & 0.00677 & 0.00606 & 0.00433 \end{bmatrix}$$

Largest = 0.00732

Sum = 0.06921 / 16 = $\boxed{0.00433 = \text{MAE}}$

⑤ $S_bad = 0.01$

quantization

$$\begin{bmatrix} 85 & -120 & 34 & 127 \\ -7 & 91 & -128 & 12 \\ 127 & 3 & -44 & -128 \\ -18 & 103 & 77 & 55 \end{bmatrix}$$

dequantization

$$\begin{bmatrix} 0.85 & -1.20 & 0.34 & 1.27 \\ -0.07 & 0.91 & -1.28 & 0.12 \\ 1.27 & 0.03 & -0.44 & -2.31 \\ -0.18 & 1.03 & 0.77 & 0.55 \end{bmatrix}$$

Sum = 2.74/16 = $\boxed{0.1713 = \text{MAE}}$

0.1713/0.00433 = 39.6

Having the S too small causes clamping and increases error significantly.